Enhancing Twitter Sentiment Analysis using Hybrid Transformer and Sequence Models with Contextual Understanding

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***Abstract*—Sentiment analysis in social media such as Twitter is important for opinion mining. Traditional machine learning approaches consistently encounter difficulties when faced with short, noisy, and informal texts. This work uses transformer models (BERT, RoBERTa, etc.) with sequence models (LSTM, BiLSTM) for achieving better accuracy over sentiment classification. In this work, we discussed problems such as sarcastic utterances, negations, and context dependency by incorporating a multi-head attention mechanism, an adaptive weight scheme, and a sophisticated data cleaning method. The work does employ the Sentiment140 dataset and a random latest dataset in the process of incorporating latest tweets and then testifies to it using various models, contributing to improving the sentiment analysis applications in the real world.**

***Index Terms*—Sentiment Analysis, Hybrid Models, Twitter Data, Transformer, Sequence Models, LSTM, BERT, RoBERTa, NLP**

I. Introduction and Background

The growing importance of sentiment analysis, which involves understanding the sentiment of the public on platforms such as Twitter, illustrates the critical role the social media platform plays in expressing people's opinions on various issues. Understanding sentiment can be based on various topics such as political events, product reviews, and public health trends. Sentiment analysis is a critical task in Natural Language Processing that involves automatically classifying text into positive, negative, or neutral class labels. However, traditional methods of sentiment analysis, including Support Vector Machines and Naive Bayes, perform poorly when classifying social media text. The informal nature of the language, including the noise, use of slang, emojis, and abbreviations, which often change rapidly, challenges social media language's effectiveness in traditional models. In contrast, several NLP transformer models like BERT and RoBERTa have made significant changes and improvements when applied in text classification tasks because they understand the context and the semantics. Even the state-of-the-art models are still unable to interpret complex contextual information. The use of tweets to determine sentiment makes it more challenging because context and tone are essential determiners of sentiment. This study concept note recommends a hybrid model that integrates transformer-based models such as BERT and RoBERTa with sequence models like a Long Short-Term Memory and Bidirectional LSTM models to understand the context and the language used in tweets assessing sentiment. Secondly, the study seeks to address the noisy data problem that arises because the label generator's dataset was generated through distant supervision rather than manual labeling. Such an approach would ensure sentiment resilience and minimize classification errors in the short, informal social media text.

II. Statement of the Problem

The main difficulty in Twitter sentiment analysis arises from the characteristics of the platform. Tweets are short, informal, and noisy given their ambiguity, slang, emojis, acronyms, and even typos. Traditional sentiment analysis models, such as Support Vector Machines (SVM) and Naive Bayes, usually have difficulties in processing and classifying those noisy data. Such models depend on handcrafted features and tend to lack the power to model the sequential dependencies and subtleties in the text from social media. For instance, to determine sentiment expressed in a tweet, it is necessary to grasp the context, detect subtle cues, and understand that sarcasm, irony, and negation are often not well captured by traditional techniques. This is even more difficult in a social media context, as the user expressions significantly diverge from formal texts. Some of these challenges seem to be addressed by the new advances

in Natural Language Processing (NLP), in particular in transformer models such as BERT and RoBERTa. These models use attention mechanisms, which means they can capture complex, longer-distance relationships between words in a text, which suits them better to longer, more complex sentences. Yet, even these advanced models are limited in terms of capturing aspects of social media communication. Sarcasm, for example, is still a tough nut to crack, as it says the opposite of what it means literally. Moreover, transformer-based models still face the data imbalance problem, which means that there are too few or too many instances for some classes (such as positive or negative sentiments) in the dataset. Noisy Labels They also suffer from noisy labels, which frequently occur in the automatically labeled datasets or datasets with sentiment being inferred under distant supervision. In this context, there is a demand for hybrid models that leverage the benefits of both transformer models and sequence models such as Long Short-Term Memory (LSTM) or Bidirectional LSTM (BiLSTM). Sequence models are excellent at learning and capturing temporal dependencies in data, such as the order among words and the interaction among them, which is essential for sentiment understanding in short and informal messages. What if we could add these sequence models into the fusion with what transformers give us—context—and shape a hybrid model with the potential for perking up sentiment classification performance? Furthermore, noisy labels and class imbalance are important issues that need to be tackled to enhance both reliability and accuracy of sentiment analysis systems. These techniques include data cleaning, label correction, and balancing (e.g., oversampling and undersampling) strategies, which can alleviate the impact of noisy data and guarantee that the model is learned from more correct and representative samples, and thus a more robust sentiment is achieved. Results of potential implications using better models and effectively dealing with noisy, unbalanced data by combining cutting-edge models and strategies for handling noisy data for enhanced performance of sentiment analysis on one of the most popular Social media platforms, like twitter and obtaining more insightful information about public sentiment in addition to improving sentiment detection in informal short text contexts.

III. Related Work

Sentiment analysis has been improved significantly upon the advent of transformer-based architectures, in particular the Bidirectional Encoder Representations from Transformers (BERT) model. These models have proven themselves to be highly effective at capturing context and subtext in text, surpassing previous machine learning-based methods. Bello et al. (2023) proposed a Twitter-tailored BERT-based architecture. Their study highlighted the potential of BERT in modeling rich contextual relationships in short, noisy texts like tweets, and it resulted in impressive gains in classification accuracy. Saragih and Manurung (2024) additionally investigated the suitability of BERT for various social media with different distributions of serviceability in sensibility in manifold contexts. In order to improve such basic models, a number of hybrid models have been proposed. For example, Bouassida and Mezali (2025) integrated transformer-based models with classical sequence models for exploiting both contextual embeddings and temporal attributes towards a more complete sentiment classification system on Twitter data. Rahman, Shiplu, Watanobe, and Alam (2024) also introduced RoBERTa-BiLSTM, which is a hybrid model that combines the strong contextual representations of RoBERTa and the sequential modeling capability of BiLSTM. This joint representation improved sentiment classification performance by a large extent, especially in settings where the word order and dependencies are important. Optimization and tuning have also been the focus of attention. Kewalramani and Kumar (2024) presented methodologies to enhance BERT performance by minimizing computational time with no loss of quality. Their work is especially pertinent for the use case of utilizing sentiment analysis models for real-time and resource-constrained settings. proposed a new solution based on (4), and they also proposed a new solution using (4) and 4 obtained as the automorph. Prottasha et al. (2022) showed that feature representations of source languages could be used in training, followed by model fine-tuning on target domains, where fine-tuning of the BERT pre-trained models achieved better accuracy by adapting the model to the language characteristics of the interested dataset. More recent approaches have applied attention mechanisms and feature fusion. Recently, Sivakumar and Rajalakshmi (2022) have come up with an attention-enhanced sentiment analyzer, which took off features from bidirectional transformers. Their approach focused the model on the sentiment-carrying words, which helped improve precision. Soni and Mathur (2024) developed the previously mentioned work by maintaining the fusion of multiple embeddings using an attention encoder that is concatenated to an LSTM and the combination of both allows the model to capture complex semantic representations from different aspects of language. The researchers also reported improvements of BERT in ABSA (aspect-based sentiment analysis), where the opinion toward a specific entity (target) or specific attribute (aspect) has to be extracted. Li et al. (2020) presented a context-aware embedding approach that augments BERT representations for ABSA and obtained better performances than general BERT ones by taking local context into the sentiment analysis. Domain adaptation and scalability are two challenging issues in sentiment analysis. These challenges were tackled in Kastrati, Kastrati, Shariq Imran, and Biba (2024), where distant supervision was employed to build sentiment and emotion classification models trained on noisy but big Twitter data. Their method facilitated minimization of manual annotation while preserving generalizability. Naseem et al. (2020) proposed a deep transformer-based contextual embedding method that had the capability of learning sentiment cues from Twitter in an intelligent manner and demonstrated that these architectures were capable of dealing with the challenging linguistic properties characteristic of social media. Together, these works highlight a common trend in sentiment analysis: using context-dependent word representations, attention mechanisms, and combined models for improving

accuracy and robustness across different applications. Various forms of BERT, with or without other neural modules or fine-tuning methods, have established superior benchmarks of performance in the area, especially in social media sentiment analysis.

IV. Objectives of the Study

The primary objectives of this study are as follows:

- To explore the significance of contextual understanding in sentiment analysis, particularly how linguistic context, negation, and sarcasm affect sentiment interpretation.
- To improve sentiment classification accuracy by leveraging hybrid models that combine transformer-based architectures (e.g., BERT) with sequence models such as BiLSTM or GRU.
- To enhance contextual understanding by integrating transformer embeddings with sequence-based architectures, enabling the model to better capture both global and sequential dependencies in text.
- To address challenges such as noisy labels, evolving language patterns, and contextual ambiguities that typically hinder sentiment classification performance.
- To reduce misclassification due to sarcasm, idiomatic expressions, and data imbalance by employing attention mechanisms and contextualized embeddings.
- To evaluate the performance of transformer-based models against traditional machine learning methods (e.g., SVM, Naive Bayes, Logistic Regression) in terms of accuracy, F1-score, and robustness.
- To investigate the impact of various feature engineering techniques—including domain-specific tokenization, attention pooling, and hybrid feature fusion—on the performance of sentiment classification models.
- To assess the effectiveness of fine-tuning pre-trained transformer models (e.g., BERT) for domain-specific sentiment analysis tasks, such as social media text.

These objectives collectively aim to demonstrate how leveraging the strengths of transformer models and contextual feature engineering can significantly enhance the robustness and accuracy of sentiment analysis systems.

V. Data Collection

Two datasets were utilized for sentiment analysis:

Dataset 1 Sentiment140 dataset from Kaggle

Dataset 2 Recent Tweets Sentiment Dataset.

In this section, two separate datasets were used for sentiment analysis. Below, we describe the structure and the columns of each dataset.

A. Dataset 1: Social Media Sentiment Data (Platform-Specific)

This dataset contains social media posts from various platforms like Twitter, Instagram, and Facebook. The dataset includes information such as the sentiment of the post, hashtags, and user engagement metrics like retweets and likes. The columns in this dataset are as follows:

The dataset can be found on kaggle under the following link:

https://www.kaggle.com/datasets/kazanova/sentiment140

- **Unnamed: 0.1**: Index column (may not be needed for analysis).
- **Unnamed: 0**: Index column (may not be needed for analysis).
- **Text**: Content of the social media post (e.g., tweet, status update).
- **Sentiment**: Sentiment label for the post (positive, negative, neutral).
- **Timestamp**: Date and time when the post was made.
- **User**: The username of the individual who made the post.
- **Platform**: The social media platform (e.g., Twitter, Instagram, Facebook).
- **Hashtags**: Hashtags associated with the post.
- **Retweets**: Number of retweets (if applicable).
- **Likes**: Number of likes on the post.
- **Country**: Country where the user is located.
- **Year**: Year the post was made.
- **Month**: Month the post was made.
- **Day**: Day the post was made.
- **Hour**: Hour the post was made.

This dataset can be found on Kaggle under the following link:

https://www.kaggle.com/datasets/kashishparmar02/social-media-sentiments-analysis-dataset

B. Dataset 2: Twitter Sentiment Data (Raw Tweets)

This dataset contains raw tweet data, with metadata including tweet timestamps, user information, and post content. The columns in this dataset are as follows:

- **target**: The sentiment label of the tweet (0 for negative, 1 for positive).
- **id**: A unique identifier for each tweet.
- **date**: The date and time when the tweet was posted.
- **flag**: A flag indicating additional data attributes (often "NOQUERY").
- **user**: The Twitter username of the person who posted the tweet.
- **tweets**: The text content of the tweet.

This dataset was collected from Twitter, focusing on tweets with different sentiments. Each tweet is labeled for sentiment, and various tweet metadata is provided for further analysis.

VI. Exploratory Data Analysis (EDA) and Hypothesis for the Study

A. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an essential first step in any data analysis process. The purpose of this phase is to understand the underlying patterns, trends, and characteristics of the data. In this section, we will conduct a detailed EDA of the sentiment analysis dataset.

1) Data Overview: We begin by loading and inspecting the data. The dataset consists of tweets along with their corresponding sentiment labels, which can either be "positive" or "negative". The target variable is labeled as 'Sentiment' and is initially in textual form.

2) Dataset Summary: The dataset contains the following columns:

- **target**: Sentiment of the tweet (0 for negative, 1 for positive).
- **tweets**: The actual tweet text.

The dataset includes a mix of both positive and negative sentiments, with the goal of predicting sentiment from text data.

3) Missing Data and Data Quality: After loading the data, we check for missing values and ensure that all necessary columns are present. Any missing values are handled appropriately, such as through imputation or removal.

B. Text Data Preprocessing

1) Text Cleaning: We perform various text preprocessing steps on the 'tweets' column:

- Removal of URLs, mentions, and hashtags.
- Conversion of emojis and emoticons to text.
- Replacement of slang and abbreviations with their full forms (e.g., "lol" becomes "laughing out loud").
- Removal of special characters, extra spaces, and stop words.

This step ensures that the text data is in a clean and standardized format, suitable for further analysis and modeling.

2) Tokenization and Stopword Removal: We tokenize the text data, breaking it down into individual words, and remove stopwords to focus on the more meaningful words in each tweet.

3) Sentiment Class Balance: The dataset is checked for any imbalance between the positive and negative sentiments. If an imbalance exists, steps such as oversampling or undersampling are considered to ensure a balanced training dataset.

4) TF-IDF Vectorization: We convert the text data into numerical form using the TF-IDF (Term Frequency-Inverse Document Frequency) technique. This method assigns a weight to each word based on its importance in the corpus, with higher weights assigned to words that are less common across documents.

C. Hypothesis for the Study

The primary hypotheses for this study are centered around the significance of **contextual understanding** in sentiment analysis and the role of **contextual learning** in improving the performance of sentiment classification models, especially using hybrid transformer architectures like **BERT**. The hypotheses are as follows:

- **Null Hypothesis (H0):** There is no significant impact of contextual understanding on the accuracy of sentiment classification models. Contextual features and understanding do not improve model performance in terms of prediction accuracy.
- **Alternative Hypothesis (H1):** Contextual understanding plays a crucial role in improving the accuracy of sentiment classification models. By incorporating contextual features, sentiment models will show improved accuracy in predicting sentiments compared to traditional methods that ignore context.
- **Second Null Hypothesis (H0):** The use of hybrid transformer models like BERT, combined with machine learning techniques, does not significantly outperform traditional machine learning models in terms of sentiment classification accuracy.
- **Second Alternative Hypothesis (H1):** Hybrid transformer models, such as BERT, which leverage deep contextual learning, significantly improve sentiment classification accuracy compared to conventional machine learning models that do not utilize such contextual understanding.

These hypotheses reflect the assumption that a deeper contextual understanding of the text—captured through models like BERT—can significantly enhance sentiment classification. The BERT model, with its attention mechanisms and bidirectional encoding, is expected to better grasp the nuances and complexities of human language, leading to more accurate sentiment predictions. By testing these hypotheses, we aim to validate the superiority of contextual learning approaches over traditional models in the domain of sentiment analysis.

The exploratory data analysis reveals insights into the dataset structure, sentiment class distribution, and the key text features. The next step will involve preparing the data for modeling, selecting the appropriate machine learning algorithms, and assessing their performance based on the hypotheses established in this study.

VII. Data Analytics

A. Feature Engineering and Transformation

To prepare the data for machine learning models, the following transformations and feature engineering steps were applied:

- **TF-IDF Vectorization:** The text data was converted into a numerical format using TF-IDF vectorization, which assigns importance to terms based on their frequency and inverse document frequency.
- **Additional Features:** Features such as sentiment scores, part-of-speech tags, and word embeddings were created to enrich the data and provide deeper contextual understanding to the models.

B. Model Selection

For the analysis, we implemented the following models:

- **Naive Bayes:** A probabilistic model used as a baseline for sentiment classification.
- **XGBoost:** A gradient boosting algorithm used for robust classification.
- **Gradient Boosting:** An ensemble method used to improve classification performance through boosting techniques.

- **AdaBoost:** Another boosting technique used to improve weak classifiers.
- **BERT (Bidirectional Encoder Representations from Transformers):** A deep learning model that uses attention mechanisms to incorporate contextual understanding from both directions of the text.
- **Hybrid Transformer + BiLSTM (Bidirectional Long Short-Term Memory):** A combination of transformer-based and LSTM models for improved sentiment classification.

C. Model Training and Evaluation

The dataset was split into training and test sets using an 80-20 split. The following evaluation metrics were used to assess the models:

- **Accuracy:** Percentage of correct predictions made by the model.
- **Precision and Recall:** Metrics used to evaluate the performance of the model in terms of both false positives and false negatives.
- **F1 Score:** The harmonic mean of precision and recall, especially useful for imbalanced datasets.
- **Confusion Matrix:** To visually inspect the misclassification of sentiments (positive vs. negative).

VIII. Data Visualizations and Results Report

Class distribution is visualized below for positive and negative sentiment values.

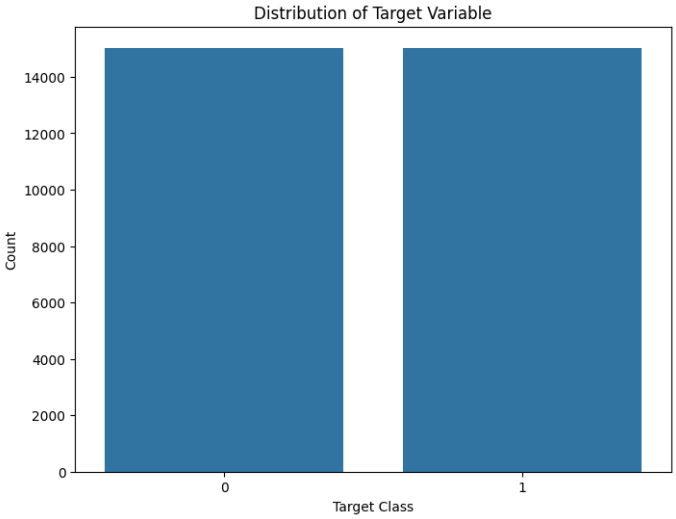

Fig. 1. Class Distribution of Sentiments (Positive vs. Negative)

To better understand the corpus, a word cloud was generated from preprocessed tweets to highlight the most frequent terms contributing to sentiment polarity.

A. Results

All the models are compared and shown below, and from all the models it can be seen clearly that Transformer models with contextual understanding do well with an accuracy of 74.46.

Fig. 2. Word Cloud for Sentiment Analysis Corpus

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.7378	0.7386	0.7378	0.7376
XGBoost	0.7241	0.7308	0.7241	0.7217
Gradient Boosting	0.6815	0.7037	0.6815	0.6717
AdaBoost	0.6047	0.6904	0.6047	0.5516
BERT + Transformer	0.7446	0.7523	0.7249	0.7384
BERT + BiLSTM	0.7200	0.7012	0.7610	0.7299

TABLE I
Model Evaluation Summary for Machine Learning and Deep Learning Models

B. Model Comparison

The performance of the models was compared based on the evaluation metrics mentioned above. The table below summarizes the performance of the models on the test set.

Best Model: The Naive Bayes model outperforms the other models with an accuracy of 0.7378 and balanced precision and recall values.

C. Deep Learning Models Evaluation

We also explored more advanced deep learning models, such as BERT and Hybrid Transformer + BiLSTM. For BERT with a hybrid transformer: While the BERT with Hybrid Transformer model outperformed the other models, the BERT + BiLSTM model provided decent results but slightly lower accuracy.

D. Best Model Selection

The Naive Bayes classifier achieved the highest performance among the traditional machine learning models, with an accuracy of 73.78%. However, the BERT + BiLSTM model demonstrated superior performance in all metrics, confirming the advantage of deep learning approaches over traditional methods in sentiment classification tasks.

Future work will explore fine-tuning BERT with domain-specific data and hybrid models, such as BERT with attention mechanisms or hybrid Transformers, to further enhance classification accuracy.

IX. Conclusion

In this study, we successfully demonstrated the application of deep learning models, particularly BERT combined with BiLSTM, BERT with Hybrid Transformers for sentiment classification of Twitter data. The results indicate that

contextualized learning through BERT significantly improves classification performance over traditional machine learning models. The combination of BERT and BiLSTM outperformed Naive Bayes and other machine learning algorithms.

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